

NATIONAL JUNIOR COLLEGE

Higher 2



CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION NO.

BIOLOGY

9747/02

JC2 Preliminary Examinations
Paper 2

22 Sept 2008

2

Hours

READ THESE INSTRUCTIONS FIRST

Write your name and registration number (e.g. 05IP0409 or 07S0620) and Biology Class (e.g. 2bi2E) clearly on the cover page.

Write your Name, Registration No. and Biology Class on the writing paper that you write the essay on.

Write in dark blue or black pen.

Answer **all** questions in **Section A** and **choose one** essay question to answer from **Section B**.

At the end of the exam, staple your essay to the back of the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINERS' USE ONLY	
1	
2	
3	
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6	
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8 / 9	
TOTAL (/100)	

This paper consists of **18** printed pages including cover page.

SECTION A: 80 MARKS

Answer all questions.

Question 1

Gastric proton/potassium-ATPase (H^+/K -ATPase) is the major enzyme involved in gastric acid secretion in the stomach. Recognizing the calming effects that tea can have on an upset stomach, biochemists quickly showed that the tannic acid found in tea is a potent inhibitor of this enzyme. Gastric proton/potassium-ATPase (H^+/K -ATPase) has two substrates, potassium ion (K^+) and ATP. (K_m is substrate concentration at $\frac{1}{2} V_{max}$)

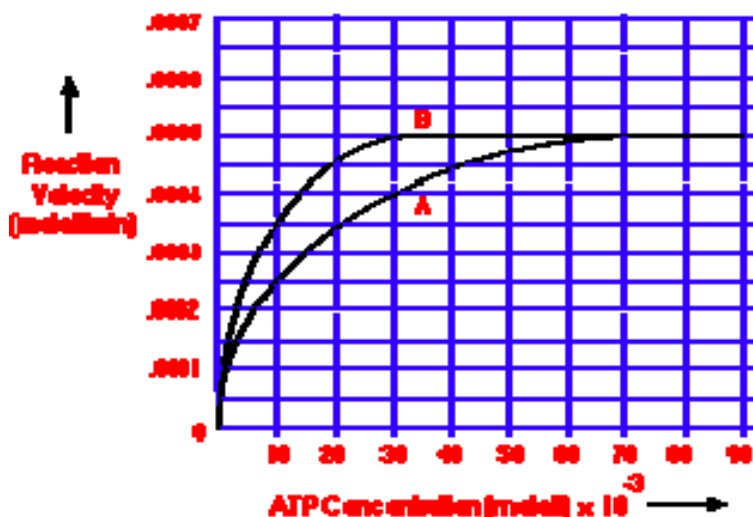


Figure 1.1 Reaction velocity-substrate concentration curves for Gastric proton/potassium-ATPase (H^+/K -ATPase) with unlimiting levels of K^+ but varying levels of ATP. A: with tannic acid; B: without tannic acid.

(a) What is the minimum ATP concentration needed to saturate the enzyme in this system without tannic acid?

30 mole/l $\times 10^{-3}$

[1]

(b) By comparing K_m values based on the graphs in figure 1, explain which treatment would result in the higher affinity of the enzyme for the substrate?

[2]

B

K_m value for B was 5 mole/l $\times 10^{-3}$

while A was 10 mole/l $\times 10^{-3}$

V_{max} would be reached at lower substrate concentrations

(c)i) Name the type of inhibition that is involved above and provide evidences for your choice.

[2]

At very high substrate concentration

V_{max} can be reached/very close to values without inhibitor added

Inhibition can be reversed/competitive inhibition

Compete with substrate for active site

(ii) With reference to the structure of the active site, suggest the possible mechanism of inhibition and why the inhibitor cannot be broken down by the enzyme.

[2]

Inhibitor is complementary in shape

To contact residues of the active site

Binding is reversible

Therefore, another substrate molecule is able to compete with the inhibitor at the active site

Catalytic residues unable to act on the bonds of the inhibitor,

Therefore cannot be broken down.

(d) Describe two ways in which enzymes interact with substrates to specifically increase the likelihood of a reaction. [2]

***Substrate binding to active site results in physical stress in bonds
Increases the likelihood of bonds breaking***

***R groups of enzymes can change the charge on the substrate/alter the distribution of
electrons/other changes
Increase the reactivity of the substrate***

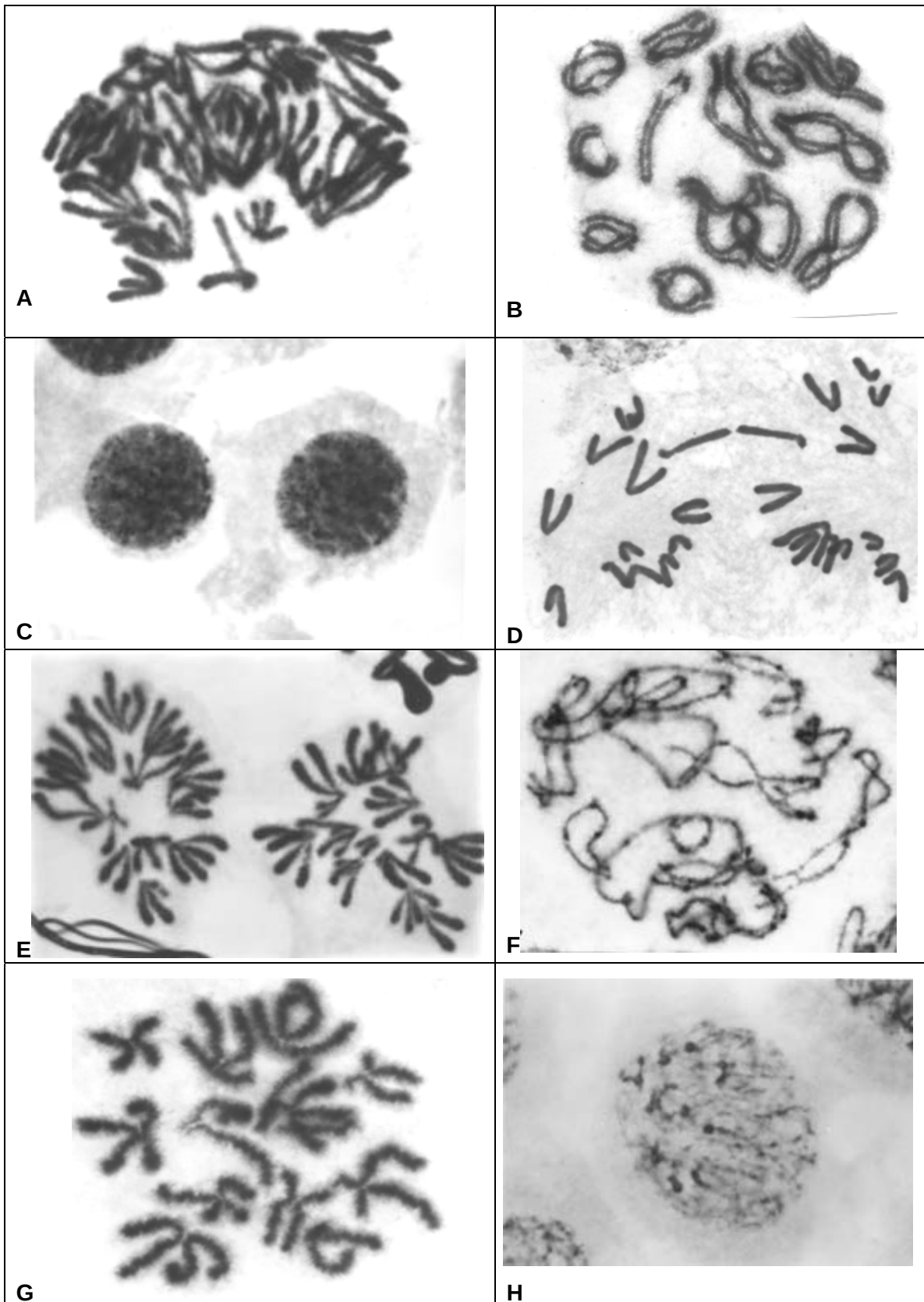
(e) Giving an example in each, describe one main difference between prosthetic groups and coenzymes that are involved in enzymatic reactions. [2]

***Prosthetic groups – tightly bonded to enzyme; Coenzyme – loosely bonded/not bonded
when reaction has not occurred
Prosthetic group: haeme group
Coenzyme: FAD, NADH***

[Total: 11 marks]

Question 2

The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant.



- (a) Micrographs **C** and **H** show the 5th and 6th stages in the sequence. In the boxes below, arrange the other letters shown on the micrographs to indicate the correct chronological sequence for this type of nuclear division. [2]

F	B	A	E	C	H	G	D
All correct						2 mk	
1 wrong e.g. B F A E CH G D						1 mk	

- (b) What is the diploid number of this plant? With reference to at least 2 stages from the micrographs, explain how you arrive at your answer. . [max 3]

Diploid number of this plant = 26
Any combination of 2 micrographs/ stages B, D, E or G, with explanations. Stage B (Prophase 1); 13 pairs of homologous chromosomes; already paired up from 26 chromosomes;
Stage G (metaphase II); Aerial view of 13 chromosomes - haploid no.; (each with 2 chromatids) lined up along equator ;
Stage D (anaphase II); 13 separated chromatids (single strands); at each pole ;
Stage E (anaphase I) ; 13 separated chromosomes at each pole; each with a centromere;

- (iv) Discuss the significance of stage B [3]

At B, prophase I
pairing up of homologous chromosomes which later allows for the alignment of chromosomes at metaphase 1.
This leads to anaphase 1, separation of homologous chromosomes and reduction in chromosome number
so that during fertilisation of gametes (restoration of diploid number.)
Non-sister chromatids of homologous chromosomes twist around each other making contact at several points along their length- chiasmata.
At each chiasma, exchange of chromosomal material between non-sis chromatids.
The breakage and rejoining between the 2 non-sister chromatids at the chiasmata results in the exchange of genes.
Leads to genetic variations among the 4 haploid cells.

Max 3

Question 3

The diagram below (Fig. 3.1) shows the process of transcription of DNA to form RNA at a particular point in time.

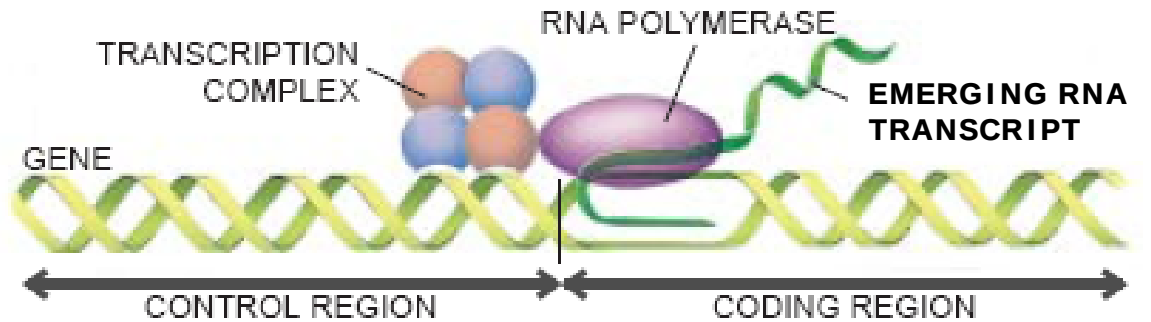


Diagram from Sci Am. Dec 1994

Fig. 3.1

a) (i) What are the main characteristics in the structure of DNA?

[4]

- ****Basic units are deoxyribonucleotides / deoxyribose sugars with a phosphate group and nitrogenous base**
- **Made up of two polynucleotide chains / made up of many deoxyribonucleotides;**
- ****forming a double helix.**
- ****Two strands run in opposite directions / they are anti-parallel.**
- **one strand runs in 5'-3' direction, the other in the 3'-5' direction;**
- ****Two strands held together by intermolecular / between the two polynucleotide chains Hydrogen bonds**
- **info about specific pairing of nitrogenous bases;**
- **Constant width of approximately 2nm / the width of a purine + pyrimidine**
- **one complete turn of the double helix has 10 base pairs**
- **one complete turn is 3.4nm.**
- **May be further super coiled or packaged with proteins**

**** points are compulsory to have. Thus if students write about other points that are true but are not these 4 main ones, they will only get 2 marks.**

(ii) Describe the events that will occur after the point in time shown in Fig. 3.1 until the completion of transcription.

[3]

(ELONGATION CONTINUES)

- **RNA polymerase continues to move along the DNA catalyzing the formation of phosphodiester bonds;**
- **adding nucleotides to the 3' end of RNA strand / synthesizing RNA in a 5'-3' direction.**
- **Complementary base pairing between DNA nucleotides and RNA nucleotides;**
- **...details on specific base pairing (DNA nucleotides base pairing to RNA nucleotides; A-U, T-A, C-G, G-C)**

(TERMINATION)

- **RNA polymerase reaches a terminator sequence;**
- **RNA polymerase, transcription complex and RNA dissociate from the DNA.**

These subsequent diagrams **b** and **c** in Fig. 3.2 and Fig. 3.3 show what happens when an artificially synthesized oligonucleotide binds to the DNA. This is a method to silence the expression of the gene.

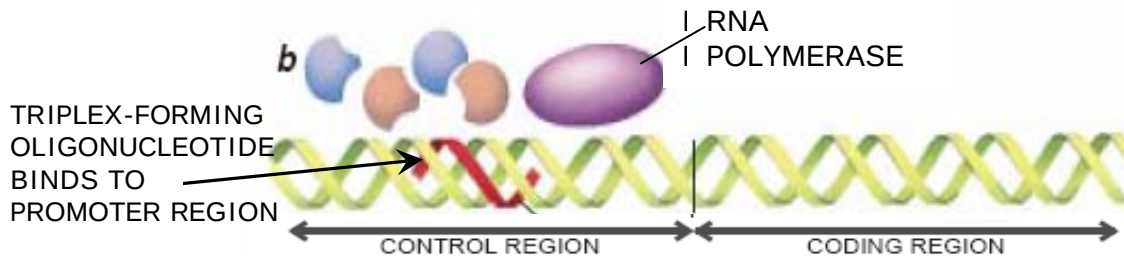


Diagram from Sci Am. Dec 1994

Fig. 3.2

b) (i) Fig. 3.2 above (Diagram **b**) shows the artificially synthesized oligonucleotide binding to the major groove of the DNA, thus forming a triple helix (triplex) in that particular region where it binds.

Explain why, according to Fig. 3.2, the RNA polymerase cannot bind to the DNA at all. [2]

- **Triple helix is no longer complementary to active site / too large to fit into the active site of RNA polymerase / Triple helix is of a different shape as compared to double helix, cannot bind to transcription complex** [1]
- **RNA polymerase or transcription complex does not bind other sequences other than those on promoter.** [1]
- **Without transcription complex binding to promoter, RNA polymerase has difficulty binding / necessity of transcription complex binding to recruit RNA polymerase** [1]

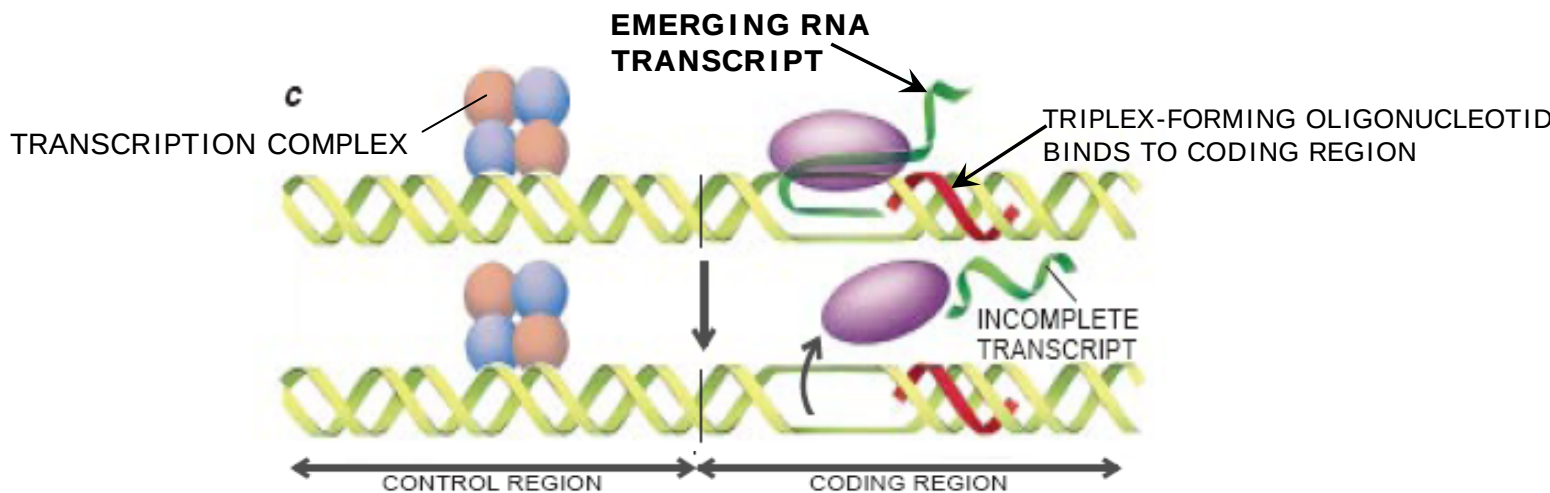


Fig. 3.3

Diagram from Sci Am. Dec 1994

(ii) Diagram **c** above shows the scenario whereby the artificially synthesized oligonucleotide binds to the coding region of the DNA. With reference to Fig. 3.2 and Fig. 3.3, account for the difference between the outcomes of Diagram **b** and **c**. [3]

- **Oligonucleotide does not bind to promoter / control region thus transcription complex and RNA polymerase can bind to promoter;**
- **to start transcription / transcription can start**
- **Thus RNA will be formed**
- **as RNA polymerase is able to move down the DNA to a certain point**

EITHER

- **But oligonucleotide disrupts the binding of RNA polymerase as it moves down / oligonucleotide causes RNA polymerase to dissociate from DNA;**
- **thus causing transcription to be terminated early / RNA formed to be incomplete or truncated.**

OR

- **Oligonucleotide prevents RNA nucleotides from being added to the RNA chain via complementary base-pairing with DNA nucleotides;**
- **thus causing RNA formed to be incomplete or truncated.**

(c) Despite the differences between the outcomes of Diagram b and c, both eventually silence the expression of the gene.

Thus briefly explain what it means to “silence the expression of the gene”. [1]

- ***To prevent production of the protein***
- ***Either by preventing production of mRNA or causing the protein produced to be non-functional.***

[Total: 13 marks]

Question 4

The diagram below shows the ears of maize (*Zea mays*). On the left, is variety Tom Thumb, the Black Mexican one on the right, and that of their F1 hybrid in the centre



A cross was made between 2 pure breeding maize varieties, Tom Thumb and Black Mexican which differed markedly in ear length. The ear length of both parents, the F1 and the F2 generations, was measured in centimeter. This is given in the table below with the number of ears in each length category (e.g.14 of the F1 plants produced ears 12 cm in length).

	Ear length (cm)																
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Black Mexican parents									3	11	12	15	26	15	10	7	2
Tom Thumb parents	4	21	24	8													
F1					1	12	12	14	17	9	4						
F2			1	10	19	26	47	73	68	68	39	25	15	9	1		

(a) Both parents showed variations in ear length. The variations in the F1 is comparable to the average of the parental variations while the variations in the F2 is greater than that found within the parental lines or the F1.

(i) Suggest an explanation for this.

[3]

Parental lines are pure breeding, therefore must be homo at all loci / genetically identical
Variations must be due effect of environment

F1 offspring should all be genetically alike/ heterozy at all loci
Variations around the mean is derived from environmental influence.

F2 should not be > susceptible to alteration by environment than parental lines nor F1

The greater variations is result not only environment effect but also genetic segregation & recombination.

(ii) Explain the genetic basis behind the wide range of ear lengths. [2]

- *Ear length is a quantitative characteristic controlled by polygenes.*
- *More than 3 pairs of independent genes.*
- *Cumulative/additive effect on the trait without complete dominance.*
- *If ear length is controlled by e.g. 3 independent loci, (Tom parent will be homozygous –aabbcc inactive alleles, and black Mexican – AABBCc active alleles)*
- *F1 fully heterozygous, AaBbCc shows intermediate ear length.*
- *In F2, the segregation & recombination of these 3 alleles pairs prod the greatest variations. E.g. AABBCc – longest.*

(iii) What is the term used to describe the range of phenotypes in the above example? [1]

Continuous variation.

(b) In another experiment, a pure-breeding pea plants with round, black seeds were crossed with pure-breeding pea plants with wrinkled, green seeds. The offspring all had round, black seeds. These seeds were grown and the resultant plants allowed to self-pollinate (fertilise).

(i) What would be the expected phenotypic ratio? Using appropriate symbols draw a genetic diagram to explain your results. [5]

Symbols:

Let B be the dominant allele for black seeds and b be the recessive allele for green seeds

Let R be the dominant allele for round seeds and r be the recessive allele for wrinkled seeds

In the cross between the parent plants:

Parental phenotype	round, black seeds	x	wrinkled, green seeds
Parental genotype		x	
(2n)	RRBB		rrbb
Gametes (n)	RB		rb
F ₁ genotype	RrBb		
F ₁ phenotype	All round, black seeds		

In the cross between F1 plants :

F ₁ phenotype	Round, black seeds	x	Round, black seeds
F ₁ genotype		x	
	RrBb		RrBb
Gametes (n)	RB rB		RB rB
	Rb rb		Rb rb

Random fertilization resulting in F₂ genotype and phenotypes:

gamete gamete	<i>RB</i>	<i>Rb</i>	<i>rB</i>	<i>rb</i>
<i>RB</i>	<i>RRBB</i> <i>round, black</i>	<i>RRBb</i> <i>round, black</i>	<i>RrBB</i> <i>round, black</i>	<i>RrBb</i> <i>round, black</i>
<i>Rb</i>	<i>RRBb</i> <i>round, black</i>	<i>RRbb</i> <i>round, green</i>	<i>RrBb</i> <i>round, black</i>	<i>Rrbb</i> <i>round, green</i>
<i>rB</i>	<i>RrBB</i> <i>round, black</i>	<i>RrBb</i> <i>round, black</i>	<i>rrBB</i> <i>wrinkled, black</i>	<i>rrBb</i> <i>wrinkled, black</i>
<i>rb</i>	<i>RrBb</i> <i>round, black</i>	<i>Rrbb</i> <i>round, green</i>	<i>rrBb</i> <i>wrinkled, black</i>	<i>rrbb</i> <i>wrinkled, green</i>

F₂ phenotypic ratio:

9 round, black seeds : 3 round, green seeds : 3 wrinkled, black seeds: 1 wrinkled, green seeds

The above cross in (b)(i) produced 324 offspring with the following characteristics.

Number of offspring	Phenotype
181	round, black seeds
51	round, green seeds
53	wrinkled, black seeds
39	wrinkled, green seeds

ii) Complete the table below and calculate the chi-square to test the significance of the differences between the observed and expected results. [2]

	round, black seeds	round, green seeds	wrinkled, black seeds	wrinkled, green seeds
Observed number (O)	181	51	53	39
Expected number (E)	182.25	60.75	60.75	20.25
Difference between observed & expected numbers (O – E)	1.25	9.75	7.75	18.75
(O – E) ²	1.56	95.06	60.06	351.56
(O – E) ² / E	0.01	1.56	0.99	17.36
$\chi^2 = \Sigma [(O - E)^2 / E]$	19.92 two decimal places.			

	Probability, p				
Degrees of freedom	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

iii) With reference to the table of probabilities (above), explain how the value for the chi-squared test supports the hypothesis that these are two pairs of segregating alleles at two loci.

At degree of freedom of 3, the calculated χ^2 value of 19.92 > critical χ^2 value (at p = 0.05 by convention) of 7.8 [2]

We reject the hypothesis that these are two pairs of segregating alleles at two loci. Ratio is not 9: 3: 3: 1. Alleles are linked and on the same chromosomes.

[Total: 15 marks]

Question 5

(a) i) State two random processes that contribute to the accumulation of allele, which do not have any benefit to survival value in a population. [2]

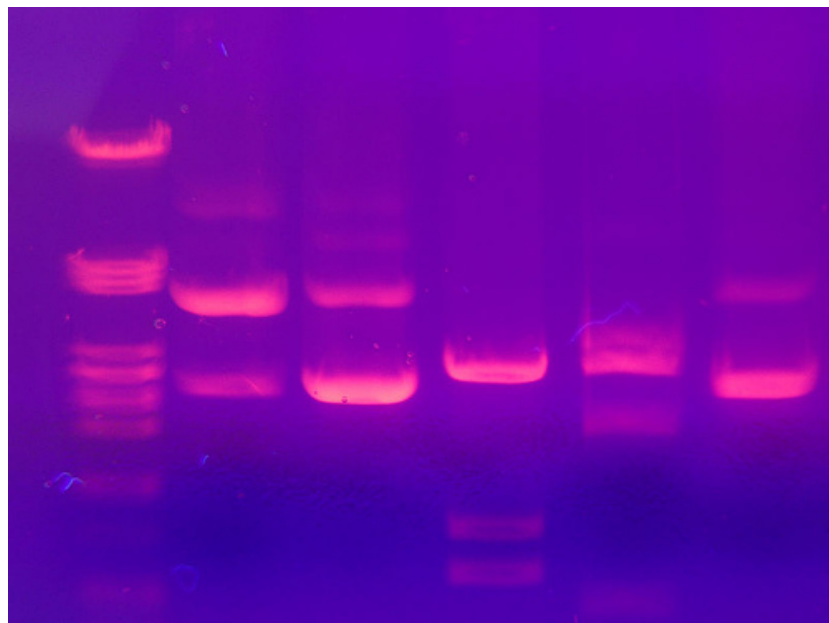
Mutation

Genetic drift

ii) Explain the basis behind the accumulation of such genes. [1]

Many mutation have little or no effect on protein function.

They accumulate because they do not affect the survival of the individual.



Mark

A

B

C

D

E

Fig shows a gel electrophoresis result of dna obtained from similar looking flagella obtained from 5 species of marine bacteria. Species A to E were each retrieved from different oceans of the world.

b (i) Name the two species that are most likely to evolve a flagella similar to A, as a result of convergent evolution. [1]

C and D

ii) Explain how this method of analysis has supported your choice of answer [2]

- **Method based on Molecular homology**
- **Chromosomes from different species**
- **Cut to produce different fragment lengths and numbers**
- **Therefore, distantly related organisms will have bands that appear on different positions of the gel.**

(c) Give four advantages of using such a method in the classification of the 5 organisms into the phylogenetic tree. [4]

- **Patterns of inheritance are straightforward, data more interpretable**
- **Molecular traits that are similar are less likely due to convergence**
- **The number of potential point mutation differences between species is likely to be very large, so even if there are independent identical mutations, those will be outweighed by the number of dissimilar mutation**
- **The number of mutations can be used to measure time because we can calculate the rate of mutation/knowning the rate of mutation, we can use the number of mutations to calculate time organism has evolved from one species to another.**

[Total: 10 marks]

Question 6

The graph below (Fig. 6.1) shows the percentage survival of mice with different genotypes at the p53 gene locus. $p53^+$ is the wild-type allele, and $p53^-$ is the mutant allele.

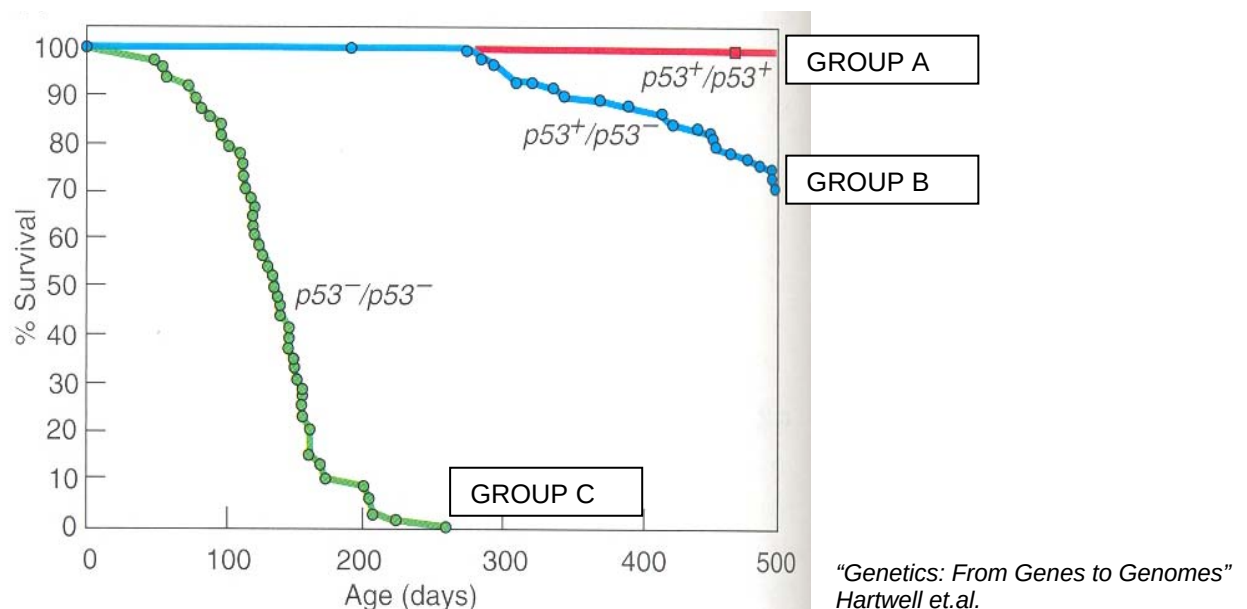


Fig. 6.1

a) Account for the difference between Group B and Group C mice. [2]

- **Group B mice have one functional copy of p53 gene**
- **while Group C have both copies of the p53 gene mutated**
- **Therefore, Group B mice still can produce some p53 tumour suppressor protein**
- **while Group C mice cannot produce any p53 tumour suppressor protein.**
- **Therefore there is a higher probability that Group C will develop tumours as compared to Group B,**
- **leading to lower rate of survival.**

b) Filling up the table below, contrast the possible genetic changes that may lead to tumour formation in the p53 gene and in a proto-oncogene. [3]

	Possible changes to p53 wild type allele	Possible changes to proto-oncogene
Point mutation in gene itself leads to:	<u>Point mutation leads to loss in function of tumour suppressor protein, decrease in activity or less resistance to degradation.</u>	<u>Point mutation leads to gain in function to give rise to oncogene increase in activity of protein or increased resistance to degradation.</u>
	<i>Must have a point for point comparison loss of fn – gain in fn etc</i>	
Point mutation in control element leads to:	<u>Point mutation in control element must lead to less transcription or no transcription / expression</u>	<u>Point mutation in control element leads to increased transcription / expression</u>
Gene amplification:	<u>Gene amplification will not cause tumours to arise</u>	<u>Gene amplification will cause tumours to arise</u>
Translocation of gene to:	<u>Translocation, gene moved to be controlled by a less active / inactive promoter may cause tumours</u>	<u>Translocation to an active promoter may result in tumours</u>

[3 complete contrasts will get full mark]

c) i) Describe the significant changes in a mouse cell from the point of acquiring of the mutant p53 alleles to the cell becoming cancerous. [4]

- **Mutant p53 alleles cannot produce functional p53 tumour suppressor proteins**
- **which are normally involved in the repair of damaged DNA or the inhibition of cell cycle (or any other eg. –check)**
- **The increased probability of mutation in DNA**
- **and the increased cell proliferation / no. of cell divisions**
- **would result in the accumulation of mutations in proto-oncogenes and tumour suppressor genes [1]**
- **For tumour suppressor genes (others besides p53), mutations must knock out both alleles to block tumour suppression**
- **Telomerase is expressed to lengthen telomeres indefinitely**
- **Allowing cell to divide uncontrollably and indefinitely, thus becoming cancerous.**

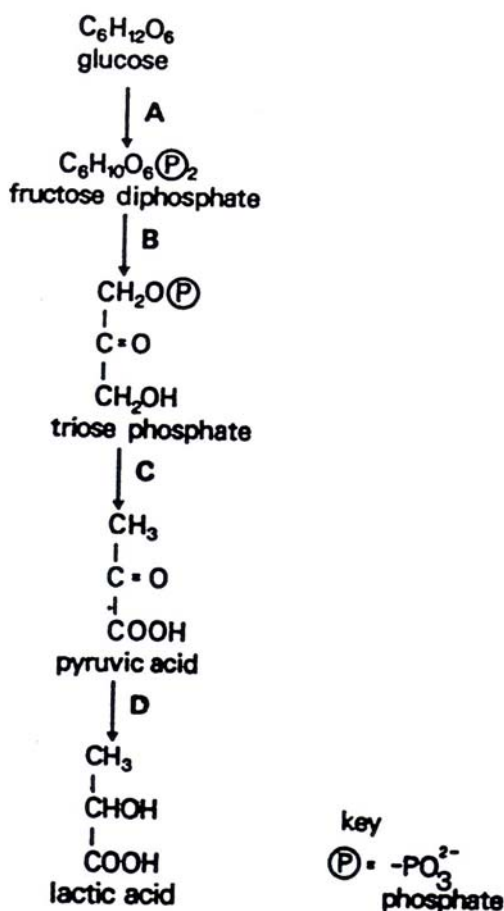
ii) How can one cancerous cell eventually lead to the death of the mouse? [3]

- **Cancerous cell divides to give more cancerous cells;**
- **forming a tumour;**
- **Tumour is malignant / cells can metastasize;**
- **and travel in the blood stream to other tissues;**
- **Disruption of function of the organ / tissue where the tumour is located [1]**
- **Diversion of nutrients from normal tissue cells to tumour cells causing normal cells to die [1]**

[Total: 12 marks]

Question 7

In anaerobic respiration in muscle tissue the conversion of glucose to lactic acid can be represented as a four stage process as follows:



(a) Name the four stages in terms of the chemical processes occurring.

- A** **Phosphorylation of glucose**
B **Cleavage / splitting (R: hydrolysis)**
C **Dehydrogenation(oxidation) & dephosphorylation(substrate level phosphorylation)**
D **Reduction / hydrogenation**

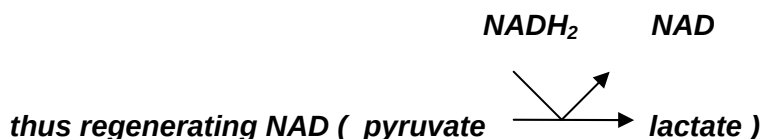
[2]

(b) At what stage would you expect hydrogen transfer to NAD to occur? **Stage C**

(c) Is there a net change in the concentrations of NAD and NADH_2 in the whole process?
 Explain your answer. [3]

No. During C, triose phosphate is oxidised / dehydrogenated
NAD acts as H acceptor, forming NADH_2 .

However, during stage D, pyruvate is reduced by accepting H from NADH_2 ,



- (d) If one molecule of glucose is converted to two molecules of lactic acid, what is the net synthesis of ATP? Explain your answer.

2 molecules of ATP.

[1]

Explanation:

During glycolysis, 4 molecules of ATP are synthesized via substrate phosphorylation from each glucose molecule converted to pyruvate acid.

However, this is offset against the 2 molecules of ATP used for activation of glucose molecule.

[2]

- (e) In what ways is the ethanol formation different from that of lactate?

<i>Ethanol formation.</i>	<i>Lactate formation.</i>
<i>In yeast cells</i>	<i>In animal cells</i>
<i>Carbon dioxide released</i>	<i>No carbon dioxide released</i>
<i>Involves 2 steps, 2 enzymes</i>	<i>Involves 1 step 1 enzyme</i> [2]

[Total: 11 marks]

SECTION B: 20 MARKS
Choose **EITHER** Question 8 **OR** Question 9.

Question 8

- a) Compare the structure and organization of prokaryotic and eukaryotic genomes. [8]
 b) Describe the various ways in which the activity of an enzyme may be controlled after a mature mRNA that codes for it has been produced in a eukaryotic cell. [6]
 c) Discuss the ways in which DNA is normally transferred from one bacterium to another without any human interference. [6]
- a) Compare the structure and organization of prokaryotic and eukaryotic genomes. [8]

Structure

Area of Comparison	Feature	Prokaryotic Genome	Eukaryotic Genome
Genes & bases	*Number of genes and bases	Larger number of bases - larger number of genes / coding regions;	Smaller number of bases - smaller number of genes /coding regions;
Chromosomes	*Number of chromosomes	Single chromosomes - usually haploid (ie only 1 set of chromosomes);	Multiple chromosomes - usually in diploid or higher ploidy levels ; (Two or more sets of chromosomes)
	*Chromosome Structure	Double-stranded circular DNA.	Double-stranded and linear DNA.
DNA Packaging	*Occurrence of DNA Supercoiling	DNA supercoiling occurs in both types of DNA	
	*Involvement of proteins in DNA packaging	Supercoiling and looping brought about by non-histone proteins , (which are positively charged).	-DNA wrapped around an octamer of basic histone proteins (also positively charged) to form nucleosomes, which are further coiled to form solenoids. -other scaffolding proteins (non-histone proteins) also involved in packaging.
	*Level of condensation of DNA	Association with histones / scaffolding proteins allow for increased structural complexity/folding to higher degree of condensation	No association with histones thus does not achieve same level of condensation complexity as eukaryote

Organisation

Area of Comparison	Feature	Prokaryotic Genome	Eukaryotic Genome
Organisation of Genomes	*Location of Functionally-Related Genes on Chromosomes	Genes that encode proteins for the same metabolic pathway are grouped together in an operon .	Genes that encode functionally related proteins are usually located on different chromosomes .
	*Control by Promoter	Genes grouped in an operon are controlled by a single promoter .	Each gene is under the control of their individual promoters .
	*Presence of Introns	Typically no introns in prokaryotic genome	Presence of introns (non-coding regions) in eukaryotic genome
	*Presence of Telomeres (can be under structure as well)	No telomeres since circular DNA allows complete replication of daughter DNA molecules.	Presence of telomeres , which are repetitive and non-coding sequences that are lost with successive rounds of replication as the chromosome shortens.
	*Location of Control Element	Control element, i.e. the operator , which activators and repressors bind to is found in close proximity to the gene it regulates .	
	*Origin of Replication	A single origin of replication exists.	Multiple origins of replication to increase the efficiency of replication of larger genome size.

[Structure: Min 3 marks, Max 6 marks]

[Organisation: Min 2 marks, Max 5 marks]

Answer needs at least 1 similarity – Reject: both made from DNA. (too simplistic) Thus maximum marks for differences: 7 marks.

- b) Describe the various ways in which the activity of an enzyme may be controlled after a mature mRNA that codes for it has been produced in a eukaryotic cell. [6]

- **Controlling mRNA stability through degradation by exonucleases**
- **Shortening of poly-A-tail to a certain length causes 5' cap to be removed**
- **and thus mRNA to be very susceptible to exonucleases**
- **Binding of micro RNAs (miRNA) complementary to mature mRNA**
- **will form double stranded RNA**
- **thus preventing ribosome from binding or moving down mRNA / translating mRNA**
- **or bring about the degradation of the mRNA**
- **Binding of regulatory proteins to untranslated region (UTR) on mRNA can also activate or inhibit translation**
- **by affecting binding of ribosome**
- **Degradation of synthesized proteins by ubiquitin**
- **determines length of time enzyme is active**
- **Modification of polypeptides to produce functional enzymes,**
- **eg. glycosylation, dimerisation, folding into specific 3D configuration**
- **Action of inhibitors**
- **competitive and non-competitive.**

c) Discuss the ways in which DNA is normally transferred from one bacterium to another without any human interference. [6]

- ***Bacteria cells can transfer DNA from one to another through the process of Transformation***
- ***whereby naked DNA / DNA fragment from one bacterium***
- ***is taken up by another bacterium through its cell wall.***
- ***DNA fragment was released into the surrounding medium***
- ***because of lysing of the first bacterium.***
- ***Following which, crossing over takes place***
- ***between the DNA fragment and the homologous segment of the bacterium chromosome, resulting in genetic exchange between DNA from 2 bacteria.***

[awarded 1x only - not v. necessary from the question: DNA transferred, not necessary incorporated into genome]

- ***Conjugation***
- ***between a donor bacterium with either Hfr or F plasmid together with a F- bacterium***
- ***is whereby a donor bacterium with either Hfr or F plasmid forms a sex pilus / conjugation bridge with a F- bacterium***
- ***DNA is replicated***
- ***and then passed through the sex pilus / conjugation bridge from the F+ or Hfr bacterium to the recipient F- bacterium.***
- ***When viruses infect bacteria, the viral progeny / resultant viruses may get bacteria DNA packaged into their capsids / virus packaging cannot differentiate between virus and bacteria fragments leading to***
- ***Transduction***
- ***occurs when such viruses infect a different bacterium, and inject the DNA from the 1st bacterium inside their capsid into the recipient bacterium.***

Maximum for only mentioning conjugation / only one method: 3 marks

Maximum without crossing over: 5 marks

Question 9

(a) With reference to the three different stages of cell signaling, describe in detail the sequence of events when a glucagons molecule reaches the liver cell. [10]

Signal reception

The hormone, glucagon acts the “first messenger”, binding specifically to a G-protein-linked receptor located on the outside surface of the plasma membrane of the target cell.

Forming a hormone receptor complex

Signal molecule is the ligand

Signal transduction

activates the G protein.

The activated G protein then activates the enzyme adenyl cyclase,

Adenylyl cyclase catalyses the formation of cAMP from ATP.

cAMP acts as a second messenger

moves within the cell to activate enzymes

such as protein kinases.

Protein kinases then activate other enzymes by phosphorylating them.

In a cascade reaction

Signal is amplified

Cellular response

regulate cellular activities in the cytoplasm

by affecting catalysis by an enzyme or cause the opening or closing of an ion channel

regulate transcription in the nucleus

by synthesis of enzymes or other proteins from genes

(b) Explain how a strong stimuli can be transmitted from the presynaptic membrane of a sensory neuron to an internuncial neuron. [7]

Process of synaptic transmission/synapse.

Electrical impulse from one neuron to another thru the release of chemicals

Arrival of action potential at the synaptic knob

Local depolarisation

Results in opening of voltage sensitive ion channels

Influx of calcium ions

Movement of synaptic vesicles toward presynaptic membrane

Fuse, rupture, release

Exocytosis

Acetylcholine/Neurotransmitters

Diffuse

Binds to specific receptors on the pre-synaptic membrane

Opening of sodium ion channels

Influx of sodium ions

Local depolarization/epsp

Exceeds threshold will initiate an AP

Strength of stimuli can be interpreted by the frequency of the AP.

(c) Explain how loss of myelination, which happens in the demyelinating disease multiple sclerosis, disrupts signal transmission in the nervous system. [3]

Myelin sheath made of lipids

is important in insulation

limiting the membrane potential to the node of Ranviers

allowing the A.P to jump from node to node

increasing the speed of transmission

without, adjacent depolarization will take place

Decreasing/slow speed of transmission